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Toward Long-Term Memory in Embodied AI: A Knowledge-Grounded 3D Scene Graph Framework

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Abstract

Advances in 3D scene understanding, particularly for embodied AI in cognitive and perceptual robotics, emphasize moving beyond object recognition toward modeling attributes and relationships that support downstream reasoning. Our previous simulator-based framework demonstrated in a 2D world that symbolic, explicit, and implicit attributes could be extracted with zero-shot detection, but its applicability to real-world 3D perception under noise and occlusion remained uncertain. This motivated our current study, where we adopt three attributes—symbolic, explicit, and implicit—building on prior work to capture both semantic meaning and physical grounding. This challenge is critical, as prior systems often allowed large language models (LLMs) to query unstructured data, producing hallucinations and inconsistent answers. Reliable reasoning instead requires anchoring LLMs to structured, knowledge-grounded scene representations. To address this, we propose a semi-automated end-to-end framework that transforms raw point clouds into a knowledge-grounded multi-attribute 3D scene graph, serving as an embodied AI memory engine for spatial intelligence. The framework contrasts hallucination-prone LLM-only querying with knowledge-grounded querying, where a semantic mediator integrates knowledge graphs to ensure fact-consistent reasoning. Experiments on diverse indoor environments show that grounding LLMs in the constructed graph reduces hallucinations, improves factual accuracy, and enables reliable semantic place inference, establishing knowledge-grounded 3D scene graphs as long-term memory engines for embodied AI and a foundation for scalable, trustworthy human–robot collaboration.

Keywords: 3D Scene Graphs, Embodied AI, Knowledge-Grounded Reasoning, Hallucination Mitigation, Long-Term Memory for Robotics

1 Introduction

True spatial intelligence in autonomous systems requires more than object recognition; it demands understanding the semantic and spatial structure of a 3D scene [1]. While open-vocabulary perception has broadened recognition [7,10,12], querying unstructured scene data with LLMs often yields hallucinations and unreliable answers [4]. As shown in Figure 1, grounding queries in a knowledge-enabled graph mitigates these issues, ensuring fact-consistent reasoning. Reliable 3D scene representations are thus essential for embodied AI in cognitive and perceptual robotics, where consistency and adaptability are critical.

To address this challenge, we present a semi-automated end-to-end framework that constructs a knowledge-grounded multi-attribute 3D scene graph serving as long-term memory for spatial intelligence. Unlike prior works that remained at a conceptual level, our method explicitly formalizes database construction and validates it in real-world environments.

A central design choice of our framework is the adoption of a human-designed tripartite ontology; symbolic, explicit, implicit attributes. This taxonomy provides the minimal yet sufficient set of dimensions for grounding both the semantic identity and physical context of objects. Symbolic attributes encode semantic labels that identify object categories, explicit attributes capture measurable physical properties such as size, color, or pose, and implicit attributes describe inter-object spatial relations. By deliberately constraining attributes to these three categories, we cover the essential aspects required for reliable reasoning while avoiding unnecessary complexity, thereby ensuring both interpretability and scalability. This bridges low-level perception with high-level reasoning,

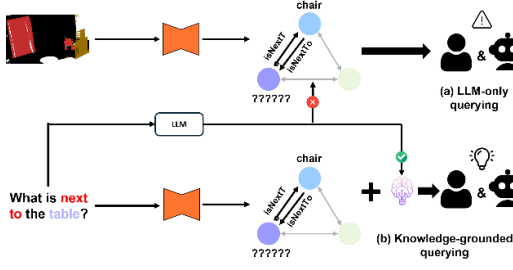


Figure 1. Knowledge-grounded querying.

(a) LLM-only querying without a semantic knowledge base often yields hallucinations and inconsistent responses; (b) knowledge-grounded querying with a semantic knowledge base grounds LLM reasoning in structured graphs, ensuring factual and reliable answers.

forming the foundation for trustworthy human-robot collaboration.

2 Methodology

Our framework transforms raw 3D point clouds into a knowledge-grounded multi-attribute 3D scene graph, functioning as an embodied AI memory engine for spatial intelligence [3,7]. The pipeline unfolds in three stages: (i) instance segmentation via Geo-Prompted SAM (GP-SAM), (ii) multi-modal attribute extraction, and (iii) semantic knowledge base generation. The overall process is illustrated in Figure 2, which shows how raw perception is progressively structured into a queryable graph representation supporting reasoning and interaction.

2.1 Instance Segmentation via Geo-Prompted SAM

Point clouds are first preprocessed to remove noise and normalize coordinates. Geometric priors such as RANSAC and HDBSCAN provide initial object-level proposals [2,3], which are then refined by Point-SAM to obtain accurate zero-shot masks [12]. The original Point-SAM employs Farthest Point Sampling (FPS) to select evenly distributed seed points and generate local patches, but this uniform sampling remains agnostic to object-level structures.

In contrast, our framework leverages geometric priors to perform **object-aware prompting**. After plane removal, we apply Euclidean clustering to group points into object candidates, and use each cluster centroid as a foreground prompt.

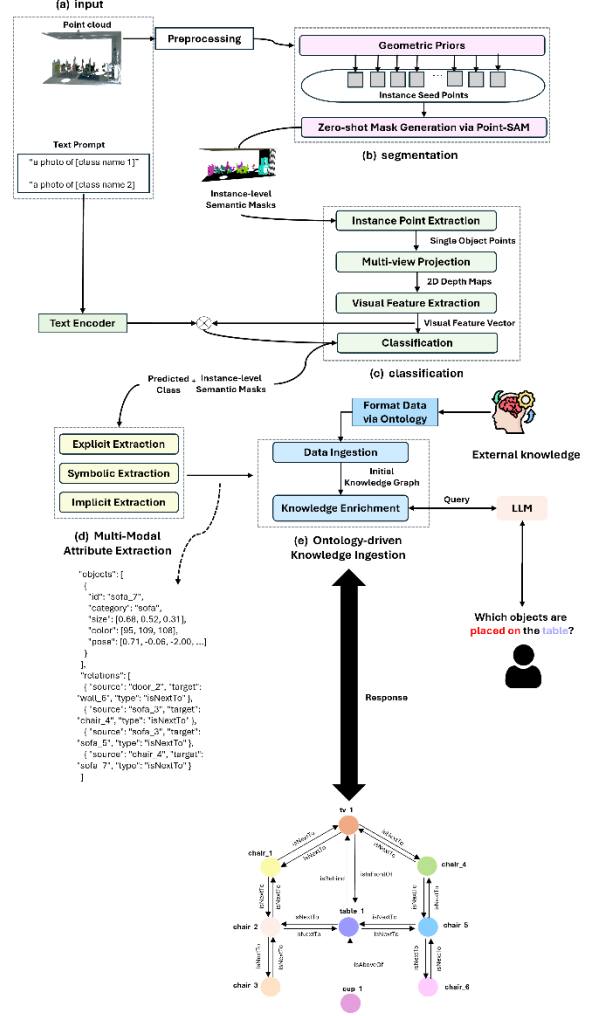


Figure 2. Overall architecture of the proposed framework.

The system transforms raw 3D point clouds into a multi-attribute scene graph through three stages—instance segmentation via Geo-Prompted SAM, multi-modal attribute extraction, and semantic knowledge base generation—ultimately enabling context-aware reasoning and interactive querying.

This ensures that prompts are explicitly aligned with object-level instances, enabling segmentation of multiple objects within the same scene while preserving the zero-shot generalization capability of Point-SAM.

Finally, multi-view projection is applied to reduce occlusion artifacts and produce more complete masks. As shown in Figure 3, our approach yields precise and complete instance segmentation across diverse environments, including cafeteria-like and kitchen-like scenes [6].

2.2 Multi-Attribute Extraction

Each segmented object is enriched with three categories of attributes, following the taxonomy adopted in [9]:

- Symbolic: semantic identity such as category or label [1,5,7].
- Explicit: measurable physical properties such as pose, size, and color [5]
- Implicit: inter-object spatial relations (e.g., isNextTo, isAboveOf, isInsideOf) inferred from geometry [2,3,5].

The resulting structured attributes are summarized in Table 1 (object node representation) and Table 2 (edge-level relations), grounding perception in both semantic and physical reality.



Figure 3. Instance segmentation results under different environments.

(a) Input Point cloud (Scene 1: cafeteria-like).

(b) Predicted instances segmentation (Scene 1).

(c) Input Point cloud (Scene 2: kitchen-like).

(d) Predicted instances segmentation (Scene 2).

Property	value	Property	value
Category	furniture	Label	Object
size	[1.2, 0.8, 0.75]	Id	table_1
Color	[150, 120, 100]	Name	table
pose	[2.1, 1.3, 0.0, 0.0, 0.0, 0.0, 1.0]		

Table 1. Example of object node representation.

Each node stores symbolic, explicit, and implicit attributes such as category, size, color, and pose, grounding perception in both semantic and physical reality.

2.3 Semantic Knowledge Base Generation

The enriched attributes are ingested into an ontology-driven semantic knowledge base, organized as a structured and queryable graph [8,11]. In our implementation, the Cogneer [8] semantic engine stores object- and relation-level attributes in a Neo4j graph database as long-term memory. During ingestion, semantic enrichment refines incomplete relations and aligns extracted knowledge with external domain priors to ensure consistency and completeness.

As illustrated in Figure 4, instance-level segmentation results generated from different environments (a, c) are transformed into corresponding semantic graphs (b, d). These graphs, which encode both object- and relation-level attributes, are then ingested into the Cogneer semantic engine and stored in Neo4j. Beyond structured storage, the knowledge base is coupled with an LLM reasoning layer: natural language queries are translated into graph operations, compositional inference is executed on factual graph data, and responses are generated in natural language form to support human-robot interaction.

This integration mitigates hallucinations while enabling grounded reasoning, interactive querying, and explainable communication between robots and humans.

Relation Type	From Node	To Node	Property
isNextTo	Table_1	Chair_7	distance = 0.45
isAboveOf	Computer_1	Table_1	Z_gap = 0.15

Table 2. Example of edge representation.

Relational links are enriched with semantic types (e.g., *isNextTo*, *isAboveOf*) as well as quantitative attributes such as distance or vertical gap, enabling semantically enriched and geometrically grounded reasoning.

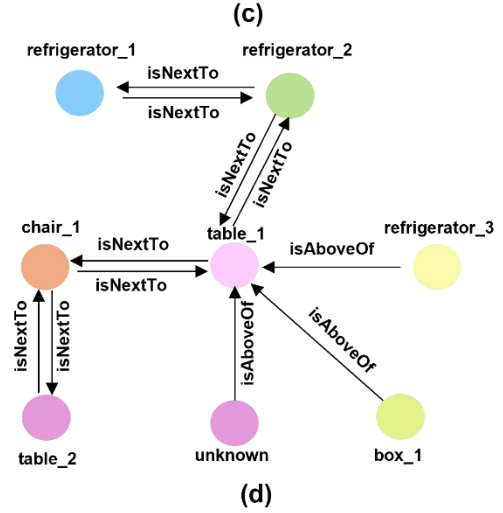
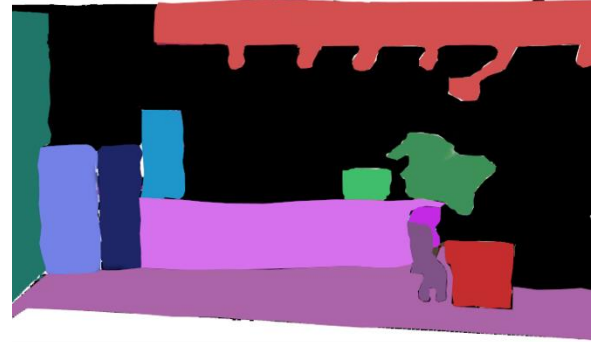
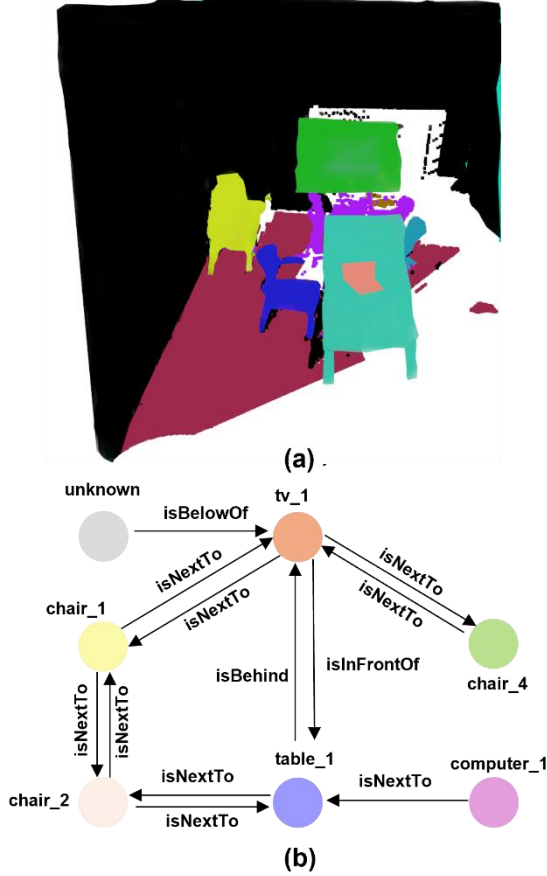


Figure 4. Semantic knowledge base generation from segmented 3D scenes.

(a), (c) showkhy040 the predicted instance segmentation masks in two different environments (Scene 1: Office-like, Scene 2: kitchen-like)

(b), (d) illustrate the corresponding semantic graphs, where nodes represent detected object inferred inter-object relationships.

These graphs are further ingested into the Cognec semantic engine and stored in Neo4j, forming an interactive and queryable knowledge base.

3 Experiment

3.1 Setup

We evaluate our framework on four representative indoor environments—cafeteria, office, corridor, and kitchen—captured with a laser scanner. These scenes ensure diverse object categories and layouts, with fixed hyperparameters to test generalization under unseen conditions.

3.2 Evaluation Protocol and Queries

We evaluate our framework against an LLM-only baseline (GPT-4) across four query types: symbolic, explicit, implicit, and scene-level inference. In total, about **160 queries** were issued across four environments. Repeated trials yielded consistent outcomes:

- **Symbolic:** 18/20 answered correctly; baseline hallucinated in 7 cases.
- **Explicit & Implicit:** more accurate attribute recognition and relational reasoning, avoiding fabricated or contradictory responses.
- **Scene-level:** 4/5 kitchen queries correct, versus only 1/5 for the baseline.

Grounding queries in a semantic knowledge base thus reduces hallucinations and improves factual consistency. Figure 5 shows two representative examples of this contrast. While full quantitative benchmarking (e.g., accuracy, precision/recall, hallucination rate) is left for future work, these results provide strong exploratory evidence of the framework’s effectiveness.

4 Results & Discussion

Figure 5 shows the contrast between unstructured and knowledge-grounded querying. With the query “*Is someone in here?*”, the knowledge-grounded setting uses Cognée’s structured scene graph to produce a fact-consistent answer, whereas the baseline relies on raw data and fabricates responses.

Across environments, the framework consistently outperformed the baseline:

- Symbolic: correctly identified object presence/absence.
- Explicit: retrieved factual attributes (e.g., color, size).
- Implicit: captured spatial relations such as adjacency.
- Scene-level: inferred environment type from detected appliances, while the baseline added spurious objects.

Overall, grounding GPT-4 in a semantic knowledge graph reduced hallucinations and improved factual consistency. A limitation is the absence of quantitative benchmarks (e.g., accuracy, precision/recall, hallucination rate). Our evaluation is qualitative but repeated trials across four environments provide

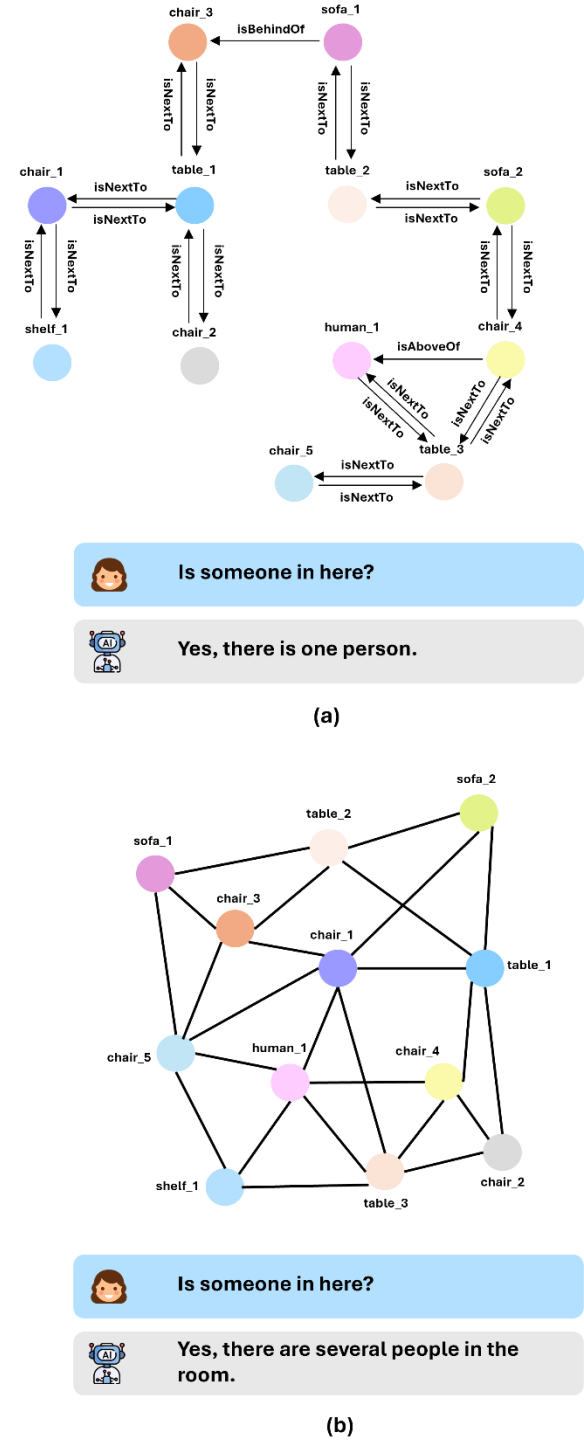


Figure 5. Comparison of query answering with and without structured grounding. (a) In the knowledge-grounded setting, Cognée provides a structured scene graph where nodes and relations are explicitly defined, enabling the LLM to give a fact-consistent answer to the query “*Is someone in here?*”. (b) In the baseline (LLM-only, GPT-4) setting, unstructured data is directly queried, leading to vague or fabricated responses.

strong exploratory evidence, with future work aimed at systematic benchmarking.

5 Conclusion

In this paper, we presented a knowledge-enabled multi-attribute 3D scene graph framework that transforms raw point clouds into structured, queryable representations by combining geometric priors, open-vocabulary segmentation (GP-SAM), and symbolic, explicit, and implicit attributes within a semantic knowledge base.

Qualitative evaluations across diverse indoor environments showed that grounding GPT-4 in the constructed graph mitigates hallucinations and enables accurate, context-aware responses across multiple query levels, validating our framework as a foundation for trustworthy human–robot interaction.

Looking ahead, we aim to scale to larger datasets, incorporate temporal dynamics, and enable real-time applications. We also envision extending the graph for richer relational encoding and actionable reasoning, allowing robots to explore, adapt, and plan tasks more effectively.

Acknowledgement

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